

series

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a sequence we will say is an ordered list of terms, which we can denote by $a_1, a_2, a_3, \dots$. a series is formed by adding the terms of a sequence: $a_1 + a_2 + a_3 + \dots$. for a finite sequence, this is simply a finite sum. for an infinite sequence, however, we cannot simply perform infinitely many additions and assume that the result is a finite number. instead, we consider the sequence of partial sums and ask whether those partial sums approach a finite limit. arithmetic and geometric sequences are particular types of sequences with simple relationships between consecutive terms.

in an arithmetic sequence, each term is obtained from the previous term by adding a constant difference, $a_{n+1} = a_n + d$, so the n th term can be written as $a_n = a + (n - 1)d$. in a geometric sequence, each term is obtained from the previous term by multiplying by a constant ratio, $a_{n+1} = ra_n$, so the n th term can be written as $a_n = ar^{n-1}$. from these terms we can form the corresponding finite series, $\sum_{k=1}^n (a + (k - 1)d)$ and $\sum_{k=1}^n ar^{k-1}$. there are other sequences that are neither arithmetic nor geometric; for now, however, we begin with these.

for a finite arithmetic series, the n th partial sum is:

1. $S_n = a + (a + d) + (a + 2d) + \dots + (a + (n - 1)d)$
2. $S_n = (a + (n - 1)d) + (a + (n - 2)d) + \dots + a$
3. $2S_n = [a + (a + (n - 1)d)] + [(a + d) + (a + (n - 2)d)] + \dots$
4. $2S_n = n(2a + (n - 1)d)$
5. $S_n = \frac{n}{2}(2a + (n - 1)d)$

for a finite geometric series, the n th partial sum is:

1. $S_n - rS_n = (a + ar + ar^2 + \dots + ar^{n-1}) - (ar + ar^2 + ar^3 + \dots + ar^n)$
2. $S_n - rS_n = a - ar^n$
3. $S_n(1 - r) = a - ar^n$
4. $S_n = \frac{a - ar^n}{1 - r} = \frac{a(1 - r^n)}{1 - r}, r \neq 1$

the restriction $r \neq 1$ comes from the algebraic expression $\frac{a(1 - r^n)}{1 - r}$, since when $r = 1$ the denominator is zero. however, the original geometric sum is still well-defined when $r = 1$. in that case every term is a , so $S_n = na$. therefore, the restriction applies to this particular form of the formula, not to the original sum itself.

let us now consider the infinite series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots$ which can be written as $\sum_{n=1}^{\infty} \frac{1}{2^{n-1}}$.

first, let us look at the series numerically by considering its partial sums:

$$S_1 = 1 = 1.0$$

$$S_2 = 1 + \frac{1}{2} = \frac{3}{2} = 1.5$$

$$S_3 = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4} = 1.75$$

$$S_4 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{15}{8} = 1.875$$

$$S_5 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{31}{16} = 1.9375$$

$$S_6 = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{63}{32} = 1.96875$$

the sequence of partial sums is therefore: $1, \frac{3}{2}, \frac{7}{4}, \frac{15}{8}, \frac{31}{16}, \frac{63}{32}, \dots$ or, numerically, $1.0, 1.5, 1.75, 1.875, 1.9375, 1.96875, \dots$

as we continue forming partial sums, we see that they appear to approach a particular value. to me it is unclear at this point if this is approaching $1.9999234532423\dots$ or 2 . maybe we should look further. perhaps it is approaching 2 , but even if we continue calculating more partial sums, I still cannot establish exactly what value it approaches.

therefore, rather than reasoning about the infinite sum directly, let us obtain S_n algebraically. once we have a general expression for the n th partial sum, we can investigate what happens to the sequence of partial sums as n approaches infinity.

using the formula for the n th partial sum of a geometric series, with $a = 1$ and $r = \frac{1}{2}$, we have

$$S_n = \frac{1(1 - (\frac{1}{2})^n)}{1 - \frac{1}{2}} = 2 \left(1 - \frac{1}{2^n}\right) = 2 - \frac{1}{2^{n-1}}.$$

we can now take the limit of the sequence of partial sums:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(2 - \frac{1}{2^{n-1}}\right) = 2.$$

so our original numerical observation was correct: the sequence of partial sums approaches 2. the infinite series therefore converges to 2. however, we can now return to our original numerical question and make it more precise. we originally asked whether the partial sums were approaching some number very close to 2 or whether they were actually approaching 2. the expression for S_n tells us exactly how far each partial sum is from 2:

$$|S_n - 2| = \left| 2 - \frac{1}{2^{n-1}} - 2 \right| = \frac{1}{2^{n-1}}.$$

therefore, the question of whether the partial sums approach 2 can be understood as the question of whether we can make $\frac{1}{2^{n-1}}$ as small as we want by choosing n sufficiently large. this gives us a way to move from our numerical intuition toward a precise definition of convergence. given any positive number ϵ , can we find an n such that

$$|S_n - 2| < \epsilon?$$

if we can, then we have not merely observed numerically that the partial sums appear to approach 2; we have established precisely what it means for the sequence of partial sums to converge to 2.